

Dose-Finding — step by step

Prepare the stocks, then fill 11 vials (nPG 4 doses + SHAM 5 doses + 2 controls). Every vial gets the same base and differs only by which 10 µL inhibitor stock goes in.

STEP 0

Master stocks from powder

Only if you don't already have them made. Both in absolute ethanol, foil-wrapped. Dissolve in ~80% of the volume, vortex hard, then top up to the exact final volume.

MASTER	WEIGH	DISSOLVE TO	= CONCENTRATION
nPG 50 mM	21.2 mg	2.00 mL EtOH	10.6 mg/mL
SHAM 250 mM	76.6 mg	2.00 mL EtOH	38.3 mg/mL

General rule if your weighed mass differs: **nPG mL = mg ÷ 10.6**, **SHAM mL = mg ÷ 38.3**. Both must go fully clear — if SHAM is slow, warm gently in the hand. These two masters feed every dilution below.

STEP 1

nPG working stocks

All four are just your 50 mM nPG master watered down with ethanol — nothing to weigh. Make ~1 mL of each, foil-wrapped. (Rule: stock = 500 × the final dose, because 10 µL goes into ~5 mL.)

WANT IN VIAL	MAKE STOCK	TAKE MASTER	+ ETHANOL
0.0125 mM	6.25 mM	125 µL	875 µL
0.025 mM	12.5 mM	250 µL	750 µL
0.05 mM	25 mM	500 µL	500 µL
0.1 mM	50 mM	master, straight	—

nPG tops out at 0.1 mM — your master is 50 mM, so you can't make anything stronger by diluting, and 0.1 mM already showed an effect last run, so the useful range is below it.

STEP 2

SHAM working stocks

Ethanol dilutions of your 250 mM SHAM master. Make ~1 mL of each, foil-wrapped.

STOCK	TAKE	+ ETHANOL	→ GIVES (FINAL IN VIAL)
6.25 mM	25 µL master	975 µL	0.0125 mM
12.5 mM	50 µL master	950 µL	0.025 mM
25 mM	100 µL master	900 µL	0.05 mM
50 mM	200 µL master	800 µL	0.1 mM

100 mM

400 μ L master600 μ L

0.2 mM

STEP 3

Two more tubes

Needed for the controls / solvent match.

- **P0** — ~0.5 mL neat DMSO (every vial gets 20 μ L of this).
- **Plain ethanol** — the two control vials get 10 μ L of this instead of an inhibitor stock.

STEP 4

Cells

Spare culture, strained, OD measured.

Make ~65 mL at OD₆₀₀ 0.0005: mL culture = 0.0005 $\times$ 65 $\div$ your culture OD, top up with YMS.

10 cell vials $\times$ 5.03 mL $\approx$ 50 mL; 65 mL leaves working margin. Blank vials use sterile YMS, not cells.

STEP 5

Fill each vial

Same base everywhere — only the last column changes.

base for every vial: 5.03 mL cells + 20 μ L P0 (DMSO) – then add the 10 μ L below

VIAL	BASE (5.03 ML + 20 ML DMSO)	ADD 10 ML OF	FINAL
A1	cells	plain ethanol	control
A2	cells	plain ethanol	control
A3	cells	6.25 mM nPG	0.0125 mM
A4	cells	12.5 mM nPG	0.025 mM
A5	cells	25 mM nPG	0.05 mM
A6	cells	50 mM nPG	0.1 mM
B1	cells	6.25 mM SHAM	0.0125 mM
B2	cells	12.5 mM SHAM	0.025 mM
B3	cells	25 mM SHAM	0.05 mM
B4	cells	50 mM SHAM	0.1 mM
B5	cells	100 mM SHAM	0.2 mM

C1	sterile YMS	50 mM nPG	blank
C2	sterile YMS	100 mM SHAM	blank

C1–C2 are optional autoxidation blanks — skip if vials are tight. All 14 still get 20 μ L P0.

STEP 6

Run

- Cap, gentle invert, no bubbles.
- Into the **27 °C** reader, dark (foil / incubator) — nPG & SHAM are light-sensitive.
- ~4 days to oxygen depletion, standard reader settings.
- Read: highest well growing like A1–A2 = the dose; ladder shape (plateau vs keeps-falling) = AOX-block vs toxicity.